Lecture 4: Adversarial Robustness (Cont.)

CS 580B1: Trustworthy Machine Learning

Fall 2024

Acknowledgments

Course slides derived from material created by Rajeev Alur and Osbert Bastani at UPenn (link to their course)

Slides from Adversarial Robustness - Theory and Practice tutorial at NeurIPS'18 by Zico Kolter and Aleksander Madry

Other relevant courses:

CS 329T at Stanford

CS 521 at UIUC

ECE1784H/CSC2559H at U Toronto

Paper we are reviewing

Towards Evaluating the Robustness of Neural Networks by Nicholas Carlini and David Wagner

Paper highlights

- Presents Carlini-Wagner (CW) attacks that are stronger than prior efforts at the time (2017)
  - Successful more frequently
  - Generate smaller perturbations

- CW attacks break the adversarial defense by “defensive distillation” methods via
  - Direct attacks against the distilled model
  - Transfer attacks, i.e., adversarial examples generated for an undistilled model are transferred to the distilled model

- Guidelines for evaluating future defenses
  - Should use these stronger attacks
  - Should demonstrate protection against transfer attacks

Paper highlights (technical)

- Targeted attacks with respect to ℓ_0, ℓ_2, and ℓ_∞ perturbations

- Objective function is
  - More suitable for gradient descent
  - Encodes box constraints using change-of-variables

- Various small but important tweaks:
  - Binary search for hyperparameter c
  - Multiple starting points for gradient descent
  - Warm-starts for gradient descent

Background: Attack algorithms

- L-BFGS
- FGSM
- Iterative Gradient Sign
- JSMA
- Deepfool

Background: Defensive distillation

- Softmax, temperature, and soft labels
- Distillation
- Defensive distillation

The three attacks

- ℓ_2 attack

$$\text{minimize} \quad \|\frac{1}{2}(\tanh(w)+1) - x\|_2^2 + c \cdot f(\frac{1}{2}(\tanh(w)+1)$$

with f defined as

$$f(x') = \max(\max\{Z(x')_i : i \neq t\} - Z(x')_t, -\kappa).$$

- ℓ_0 attack

$A \leftarrow Set\ of\ all\ pixes$
$x' \leftarrow \ell_2\ attack\ on\ A$
$while\ x' is\ a\ valid\ adv\ example$:
    $A \leftarrow A\ - Least\ relevant\ pixel$
    $x' \leftarrow \ell_2\ attack\ on\ A$
$return\ (x')$

- ℓ_∞ attack

$$\text{minimize} \quad c \cdot f(x+\delta) + \cdot \sum_i \left[(\delta_i - \tau)^+\right]$$

Attack evaluation

- Different combinations of objective functions and box constraint encodings

- CW attacks vs previous attacks on undefended (i.e., undistilled) models

- CW attacks on distilled models

- Transferability of adversarial examples generated via CW attacks

Discussion questions

- What specific characteristics of the new attack algorithms contribute to their higher effectiveness compared to previous methods?

- Why does CW attack succeed in bypassing the distillation defense?

- Why did distillation help against attacks prior to CW?

Discussion questions

- Why does the paper focus on the $\ell_0, \ell_2, \ell_\infty$ distance metrics, and are there other metrics that could provide additional insights into adversarial robustness?

- How to design a good objective function?

- Why do attacks transfer?

- Is there a universal defense against **ALL** adversarial attacks? Alternately, how to design stronger defenses? Potential for adversarial training as a defense mechanism

Discussion questions

- Computational efficiency
- Role of temperature
- Difference between attacking images and texts
- Would a quantum computer be able to find adversarial examples?
- Ethical considerations of publishing such attacks

Discussion questions

- Connections to model steering
- Can the CW attacks be applied in a black-box setting?